Guided by Primes

Decomposing Problems into Reusable Atomic Spans

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Mycelium is the underground root network of fungi. It decomposes organic matter into simple molecules and distributes nutrients across entire forests. This project does the same for math problems by decomposing them into atomic spans.

Abstract

We present Mycelium, a system for solving math word problems by distilling attention patterns from large language models into lightweight classifiers. The key insight is that transformer attention reveals semantic spans that form cohesive operations or sub-graphs. We extract these patterns from Qwen 7b, train a mapping to predict them on MiniLM 22m, and run inference with MiniLM with ~200 specialized templates. Our approach achieves ~95% correlation with the teacher model's attention patterns while being 318X smaller for GSM8K.

Introduction

Math word problems require decomposing natural language into executable operations. The challenge is identifying semantic spans — multi-token units that represent single operations. "Half the price of the cheese" is one operation (multiply by 0.5).

We call this the Panama Hats Problem: just as "panama" means country but "panama hats" means something entirely different, mathematical phrases have compositional meaning that naive tokenization destroys.

Large language models implicitly learn these span boundaries through attention patterns. Tokens within a semantic span attend strongly to each other. We extract this signal and distill it into a fast, lightweight system.

Attention Signals

We extract three signals from attention matrices:

Attention Entropy (per token)
- Low entropy → token attends to specific targets → important structural role
- High entropy → token attends broadly → less discriminative

Attention Received (per token)
- Sum of attention each token receives from all other tokens
- High received → many tokens look back to this one → entity or anchor

Span Connectivity (per span)
- Average mutual attention between tokens in a candidate span
- High connectivity → tokens form cohesive unit → valid span
- Low connectivity → tokens don't belong together → split or reject

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Why MiniLM is Perfect for Distillation

MiniLM was originally trained with: loss = MSE(student_attention, teacher_attention)

This means MiniLM already learned to mimic attention patterns from a larger teacher. When we fine-tune it on Qwen 7B attention patterns, it's doing exactly what it was designed for but with a new teacher.

Key advantages:
- Bidirectional encoder → sees full context
- Sentence-transformer architecture already optimized for semantic similarity
- The training objective aligns perfectly with our goal

Trained Signal Mapping

We process the GSM8K dataset with our “panama hats” algorithm on both Qwen7B and MiniLM which produces ~17k spans with both MiniLM embeddings and Qwen attention signals that we use to train a mapping:

Training Pipeline:
1. Extract Qwen 7B and MiniLM attention patterns on GSM8K
2. Creates ~17k spans (“apples”, “cookies”)
3. Change to generic {entity} then deduplicate 17k spans → 200 spans
4. Embed ~200 spans now sub-graphs or specialized templates
5. Build custom DSL for each ~200 spans
6. Train mapping: predict Qwen signals from MiniLM features

Cross-Attention Between Spans

In addition to spans we track:
1. Sequence awareness — Position in the problem
2. Previous span tracking — What operation came before
3. Entity tracking — Which entities have been introduced and referenced

Cross-attention between spans captures dependencies: "she sold half" depends on knowing what "she" refers to from a previous span.

Inference Pipeline

1. Run MiniLM 22M params
2. Apply learned mapping → approximate Qwen signals
3. Embedding and attention signals for span detection + template matching
4. LLM executes specialized template

No Qwen 7B needed at inference, just ~200 spans + trained mapping + MiniLM

Specialized Templates with Generic Entities

Each span maps to a specialized template, and the LLM executes the DSL. These templates encode the mathematical structure of common operations: circle geometry, ratios, percentages etc.

We use generic entity placeholders, instead of apples, cookies etc. from the GSM8K dataset, but the underlying operations are identical. "Half the apples" and "half the cookies" both map to the same template. By replacing specific nouns with {entity} placeholders, we get specialized structure with generic applicability.

Building the Graph

Deduplicated Spans = Specialized Templates = Sub-Graphs

Steps
- Match MiniLM spans to specialized span templates
- Granularity – span level; guided by our “panama hats” problem
- Sub-graph composition - guided by attention span connections

Once the graph is built, we topologically sort and execute each node in order. Each template's DSL runs with its extracted values. The final node's output is the answer.

Results

End-to-end test on held-out samples shows 96.8% average correlation — better than training (94.5%).

Aren’t you overfitting – yes we kind of are and that’s okay.

Conclusion

We demonstrated that:
1. Transformer attention patterns reveal semantic spans in math problems
2. These patterns can be distilled from Qwen 7B to MiniLM → 318x smaller
3. MiniLM's original training objective (MSE on attention) makes it ideal for this task
4. Cross-attention between spans captures sequence and entity dependencies

5. Specialized templates with generic entities enable execution
6. Span templates compose into computation graphs via attention connectivity

The key insight: large models learn span boundaries implicitly. We make this explicit through attention distillation, enabling fast inference with an LLM executing structured templates.

Acknowledgments

Built with Claude Code. The velocity of iteration was extraordinary — Claude enabled thinking at a higher level of abstraction, focusing on system design rather than implementation details.

On the primes analogy. Primes = Spans = Templates = Sub-Graphs: This project was like a choose your own adventure book where Claude was the narrator asking questions about direction and I was the reader. We would go back and forth on implementation details, questions would come up, and my answers were largely guided by primes. The analogy was so strong, it made many decisions easy.

The Quiet Singularity and Saddle Crossings

A Pringle chip is a saddle crossing or hyperbolic paraboloid curving upward along one axis and down along the perpendicular axis. At the center of the chip you're at maximum height in one direction and minimum height in the other.

Saddle crossings don't announce themselves. You could pass through one without fireworks only noticing later that the dynamics have different character. The pre and post distinction is real but the transition can be unremarkable. The singularity might be quiet.

The singularity isn't a moment where everything accelerates. It's a moment of extreme sensitivity to which unstable manifold we depart along. Small perturbations now mean dramatically different basins of attraction later.

On the Pringle, whether you're ascending or descending depends entirely on which direction you're moving, suggesting the singularity might be accelerating or decelerating or both simultaneously along different axes. Economic concentration is accelerating while coherence of institutions declines and capability of small groups rises.

A Pringle has one unstable and one stable direction. But in a high dimensional system, you could have a saddle with many unstable directions. The singularity might be a high index saddle with lots of ways to fall off and a few stable ridges to walk.

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